How long should I boil my egg?

A worked example of an evidence ledger*

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Abstract

A hard-boiled egg is done when its centre reaches 85 °C. Starting from the cooking-time formula of Buay et al., we show that bigger eggs need quadratically longer, that colder eggs need longer, and that an egg taken from the fridge needs about one minute more than one at room temperature, however hard it should be. A kitchen measurement agrees. This paper is an example for the trust-ledger tool: the kitchen, its data and its authors are fictitious, and the appendix shows what the ledger records about each result.

1 The model

An egg is a prolate spheroid with semi-axes $a > b$. For heat flow, Buay et al. replace it by the sphere with the same ratio of surface to volume, of radius

$$R = \frac{2ab}{b + \beta a}, \qquad \beta = \frac{\arcsin e}{e}, \quad e = \sqrt{1 - b^2/a^2} \tag{1}$$

[1, eq. (22)], and they keep only the first term of the series solution for the temperature at its centre. We take their result as given.

Lemma 1 (Buay et al.)**.** *A sphere of radius R and thermal diffusivity α, at temperature T_0, put into a bath at T_b, reaches the temperature T_c at its centre, $T_0 < T_c < T_b$, after the time*

$$t(R, \alpha, T_b, T_0, T_c) = \frac{R^2}{\pi^2 \alpha} \ln \frac{2(T_b - T_0)}{T_b - T_c}.$$

The formula is their eq. (11). It is an approximation (the first term of a series), best when the egg is nearly done (T_c close to T_b) and worst for a very soft egg; they compare it with a measurement and find agreement within a few percent at $T_c = 85\,^\circ\mathrm{C}$ [1]. An older and simpler estimate [2] is about 15 % too long.

2 Three consequences

Lemma 2 (Bigger eggs)**.** *For every k, $t(kR, \alpha, T_b, T_0, T_c) = k^2\, t(R, \alpha, T_b, T_0, T_c)$.*

Lemma 3 (Colder eggs)**.** *If $R, \alpha > 0$ and $T_0' < T_0 < T_c < T_b$, then*

$$t(R, \alpha, T_b, T_0, T_c) < t(R, \alpha, T_b, T_0', T_c)\,:$$

a colder egg needs strictly longer.

*Evidence ledger root `66c3b751401c`; see the section "Evidence ledger" at the end of this paper.

Theorem 4 (The fridge penalty)**.** *Assume Lemma 1. If $R, \alpha > 0$ and $T_0' < T_0 < T_c < T_b$, an egg started at T_0' needs longer than one started at T_0 by*

$$\frac{R^2}{\pi^2 \alpha} \ln \frac{T_b - T_0'}{T_b - T_0},$$

which does not depend on T_c.

For a medium egg ($R = 23.2\,\mathrm{mm}$) and $\alpha = 1.6 \times 10^{-7}\,\mathrm{m^2/s}$ the prefactor is $R^2/(\pi^2\alpha) = 341\,\mathrm{s}$. In boiling water, an egg from the fridge (5 °C) instead of from the counter (20 °C) needs $341\,\mathrm{s} \cdot \ln(95/80) \approx 59\,\mathrm{s}$ more: one minute, for a soft egg as for a hard one.

Lemma 5 (On a mountain)**.** *Where water boils below 100 °C, every egg takes longer.*

This lemma follows from the same formula, but we did not put it into the ledger: the appendix lists it as not covered, and nothing is claimed about it there.

3 In the kitchen

We measured one egg: its semi-axes with a calliper, then its centre temperature with a probe thermometer every 30 s after putting it, straight from the fridge, into boiling water. A script fits α to the readings with the full series solution [1, eq. (5)].

Observation 6. *Our egg ($a = 28.0\,mm$, $b = 21.5\,mm$, so $R = 23.2\,mm$ by (1)), started at 5 °C, reached 85 °C at its centre after 14.4 min. The fitted thermal diffusivity is $\alpha = 1.60 \times 10^{-7}\,\mathrm{m^2/s}$, inside the range 1.5 to $1.8 \times 10^{-7}\,\mathrm{m^2/s}$ found by Buay et al.*

Three people ate the egg. In a real paper their notes would stay private, and the ledger would hold only a salted commitment to them, so that they cannot be changed later without anyone noticing. In this example the notes and the salt are in the supplement, so that the reader can check the commitment.

Observation 7. *All three tasters called the egg firm but not rubbery, and none saw a green rim around the yolk.*

Conjecture 8. *Six minutes from boiling water makes the perfect soft-boiled egg.*

4 What the reader can check

What the ledger says. An evidence ledger records what each of 6 results of this paper rests on (by evidence type, checked proof: 3, numerical evidence: 1, assertion: 2). Machine checks: Lean for 3, Check for 1; 1 of the results holds given Lemma 1, which the formal proofs assume and do not prove, and which A. Cook attests to be a known theorem of the cited literature. 2 have no machine check. Lemma 5 is not in the ledger. 5 of the 6 results are attested by A. Cook, B. Taster and C. Reader (4 green, 1 conjecture), 6 of 9 attestations signed. 1 has only attestations made before its statement last changed. The ledger is sealed by its root hash `66c3b751401c…`[1], which covers the evidence, the statements as typeset here and the source of this paper; the section "Evidence ledger" at the end of this paper lists every result, explains the attestation levels and how to check the root.

[1] SHA-256 `66c3b751401ca991a41b19bce0396362d68602d2f04bbeff85a0e5d96028eda2`.

References

[1] D. Buay, S. K. Foong, D. Kiang, L. Kuppan, and V. H. Liew. How long does it take to boil an egg? Revisited. *European Journal of Physics*, 27:119–131, 2006.

[2] P. Roura, J. Fort, and J. Saurina. How long does it take to boil an egg? A simple approach to the energy transfer equation. *European Journal of Physics*, 21:95–100, 2000.

Evidence ledger

Status. Of 6 results in the ledger: 4 **GREEN**, 0 **YELLOW**, 1 **CONJECTURE** (a different kind of statement, not a worse grade); 1 with only **STALE** attestations; 0 not yet attested. Machine checks: 1 hold given a hypothesis that is assumed, not proved. 6 of 9 attestations carry a valid signature.

Root hash (SHA-256, sealed 2026-10-19, ledger format 0.3.0):

```
66c3b751401ca991
a41b19bce0396362
d68602d2f04bbeff
85a0e5d96028eda2
```

The root is a Merkle hash over the evidence behind each result as recorded by the tools that checked it (Lean, Check), the data and processing steps it rests on, the statements of the results as typeset in this paper, the assumptions and cited hypotheses, the assertions made by the authors, the attestations given by named persons, the environment (tool versions), and the source of this paper (hash `cb7325105d87`, excluding the files that show the root and this table). Changing any of these changes the root. Supplementary data (ledger files, sources, verification tool): the directory examples/egg of the trust-ledger repository.

What it does and does not establish. The root shows that the data in the supplement is exactly the data this table was derived from. It does not show that a formal statement means what the text means; that is what attestations are for. An attestation records a person, a date and the checklist items that person did; it is bound to the hash of the statement as typeset, and is shown as **STALE** when the statement has changed since. GREEN: the named author has personally done the checklist and stands behind the result. YELLOW: convinced and the gap repairable, but would not yet defend it against experts. CONJECTURE: a claim without proof. The tool columns are machine checks and are shown in a neutral style.

Attestations do not propagate. A green attestation on a result says that the attester has checked that the formal statement means what the paper's statement means. The proof, including every lemma it uses, is what the tool checked. A lemma attested only yellow therefore does not weaken a result whose formal proof uses it; its level concerns the lemma's own statement in the text. What does carry over is listed with the result: the hypotheses it is proved given, and evidence outside the tool's proof check.

How to check. Recompute all hashes from the supplement and compare with the root above: `python3 ../../ledger.py -P egg verify`. To also re-derive what the tools recorded, rerun them from the sources in the supplement (`ledger.py extract`, `ledger.py check`; see its README), then run the same command.

Result	Ev.	Stmt.	Lean	Check	Reg.	Attestation
Lemma 2 `lem:scaling`	E1	tex	checked	—	0	**GREEN** 2026-10-14 A. Cook [signed]
						GREEN 2026-10-14 C. Reader [signed]
Lemma 3 `lem:colder`	E1	tex	checked	—	0	**GREEN** 2026-10-14 **STALE** B. Taster
Theorem 4 `thm:fridge`	E1	tex	checked, given [H1]	—	2	**GREEN** 2026-10-14 A. Cook [signed]
						YELLOW 2026-10-14 B. Taster
Observation 6 `obs:kitchen`	E4	tex	—	pass	2	**GREEN** 2026-10-19 A. Cook [signed]
Observation 7 `obs:taste`	E5	tex	—	—	1	**GREEN** 2026-10-14 B. Taster
Conjecture 8 `conj:six`	E5	tex	—	—	2	**CONJECTURE** 2026-10-14 A. Cook [signed]

Hypotheses. Results marked *given* [H1] are shown by a tool given the hypothesis H1 of the table below, a statement that is assumed there, not proved. If it is a known theorem of the literature, a named person can attest that: they found the cited statement, compared it with the formal definition and checked that the conditions match. The attestation is bound to the hash of the statement in this paper, the formal definition and the cited sources, so it becomes **STALE** when any of them changes. It does not make the hypothesis a machine-checked proof; it says who vouches that the assumption is a theorem. Currently 1 of 1 hypotheses are attested as known theorems.

Key	In this paper	Formal hypothesis	Cited	Attested as known theorem
H1	Lemma 1	`Egg.FirstOrder`	buay	**GREEN** 2026-10-14 A. Cook [signed]

Not in the ledger. 1 labelled theorem-like results of this paper have no entry in the ledger, so nothing is claimed about them here: Lemma 5 `lem:altitude`.

Keys. Signatures are optional and use OpenSSH keys. A key is introduced by this paper (the authors' keys), by vouching, or by a recorded source such as a homepage; there is no certificate authority, so the binding of a key to a person rests on this list being published under the authors' names.

Person	Fingerprint	Introduced by
A. Cook	`SHA256:cGJtCMtPh+hNf/NcNSbbXVSzsHwoQx21foejWLDcuu8`	paper
C. Reader	`SHA256:JdKJKJPweu8XmWsJ1NA+11ml2EmjLRx28TNcCOCQvzw`	attestation-file

Checklist. An attestation lists the items of this checklist that the person has done: C1: compared the statement in the paper with the formal statement or check it rests on; C2: re-ran the machine check myself; C3: checked the numbers and units in the text; H1: found the cited statement in the source; H2: compared it with the formal definition; H3: checked that its conditions match the use here.

Ev.: evidence type, E1 checked proof, E2 checked certificate, E3 exact computation, E4 numerical evidence, E5 assertion. Stmt.: `tex` = taken from this paper's source by its label, `manual` = written in the ledger. Lean: checked = accepted by the Lean kernel with the standard axioms only; checked, given [Hk] = in addition the hypothesis Hk of the Hypotheses table is assumed, not proved; extra axioms = the proof uses an axiom beyond the standard ones; contains sorry = the proof is unfinished. Check: pass/fail = the check script, which tests the result against its data, exits with 0 / another code on the recorded inputs. — = no evidence of that kind. Reg.: number of open assumptions

on which the result rests (assumed hypotheses, cited results, data taken as given, manual steps, assertions). A result covering several labels appears once per label.